

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI (PILANI CAMPUS)
FIRST SEMESTER 2015 – 2016

Course: CE F214 Construction Materials
Component: Comprehensive Exam (Closed Book)

Date: 07-12-2015 (9:00 - 12:00 FN)
Max. Marks: 95

Name:

ID No:

Part A: Short Answers

[20 x 2 = 40]

1. Write Bogue's compounds?
2. List four important raw materials used in cement production
3. Write structure of any three cement hydrated products
4. Draw heat of hydration pattern for cement and specify different stages.
5. Explain purpose of brick masonry in framed and load bearing structures.
6. Write three commonly used fibres in FRP applications
7. Draw stress-strain diagram of mild steel and HYSD
8. Differentiate: Steel, Iron and stainless steel
9. Explain: ITZ
10. Explain basic principle of XRF spectroscopy.
11. What is known as Efflorescence?
12. What is difference between porosity and permeability?
13. What is known as Glulam?
14. List two major decays of timber
15. Explain electrostatic repulsion of superplasticizer.
16. Write any two major advantages of glass compared to other construction materials.
17. What is known as pigment volume concentration
18. How can you find ductility of bitumen as per IS?

19. Explain: Holidays in steel reinforcement

20. Explain: PAI

Part B

[55 Marks]

1. Explain the following patterns in brick masonry with neat sketch.

English bond (2)

Flemish bond (2)

Stretcher bond (2)

Explain: Procedure of efflorescence test for brick as per IS 3495-Part 3. (2)

2. Explain the followings with a neat sketch

a) Macroscopic structure of wood (2)

b) Microscopic structure of wood (2)

c) Influence of moisture content on compressive strength of timber (3)

3. What is known as cement-admixture compatibility? (1)

List four major factors influence on cement-admixture compatibility (2)

Explain Marsh cone and mini slump test methods (4)

List all test methods used to determine workability and durability of concrete (3)

4. List different types of steel reinforcement (2)

Write a brief note on properties, surface and core structure of TMT steel reinforcement (3)

Short note on stainless steel reinforcement (2)

Explain Re-bend test for steel reinforcement (2)

5. Write test procedure of following tests (6)

Compaction factor test

Softening point test for bitumen

Impact test for aggregate

6. What is pozzolanic reactions? (2)

Explain: Pore refinement process (2)

List any six pozzolanic materials. (2)

List any eight special concretes. (2)

7. Write step-by-step procedure of Mix design of concrete as per IS 10262-2009. All important specifications need to highlight in the procedure. (7)
